

AIMLCZG546 - Software Engineering for Machine Learning

Assignment II

Overview

This assignment builds on Assignment I and focuses on the implementation, code quality, and testing/QA aspects of building an ML-enabled system. You will take an ML application (**either the same one from Assignment I or a new one**, per the problem statement chosen) and demonstrate sound coding practices, refactoring, API design, and a rigorous testing strategy — including deployment-readiness aspects where applicable.

General Instructions

- It is a Group Assignment. Please follow the naming convention as <Group no>.ipynb.
- Inside each report and implementation notebooks, you are required to mention your name, Group details.

Group Details

Group No: <Fill Group No>

Group Member Names with Contribution

Sl. No	BITS ID	Name	Contribution of team member (Qualitative)	Percentage Contribution (Out of 100) Quantitative
1				
2				
3				
4				

- Weightage - 10 marks
- Assignment 2 Submission Date – 15th August 2026, 23:00.
- Submissions after the due date will result in deduction of marks.

Note

1. No extension of deadline will be given under any circumstances. Manage your time and complete the assignment within the deadline.
2. Since Assignment-II is part of EC-1 component, there is NO MAKEUP for this component.
3. Any questions or clarifications regarding the assignment can be emailed to "omshree.b@wilp.bits-pilani.ac.in".

Project Details

Objective 1: Implementation and Code Sharing [5 Marks]

1. Refactor/design your ML application code using appropriate OOP and/or functional programming principles (e.g., separate classes/modules for data ingestion, feature engineering, model training, and inference).
2. Clearly distinguish and demonstrate the difference between research code vs production code for at least one component (e.g., a Jupyter notebook prototype vs. a modularized, production-style script/package).
3. Implement proper error handling and logging (e.g., using Python's logging module) across at least 3 critical functions/modules, with meaningful log levels (INFO, WARNING, ERROR).
4. Apply code formatting and linting tools (e.g., black, flake8/pylint, isort) to your codebase; include before/after screenshots or lint reports.
5. Design and implement a simple REST API (e.g., using FastAPI) to expose the model's inference functionality, following good API design practices (clear endpoints, request/response schemas, status codes).

Objective 2: Quality Assurance [5 Marks]

6. Identify and implement at least 2 different types of tests for your ML system (e.g., unit tests, integration tests, data validation tests) using pytest.
7. Implement specific tests for ML components:
 - a. Testing model training (e.g., overfitting on a small batch, checking loss decreases)
 - b. Testing model inference (e.g., output shape/range checks, invariance/directional tests)
8. Define and measure at least 2 metrics each for:
 - a. Model Quality (e.g., accuracy, F1, calibration)
 - b. Data Quality (e.g., schema validation, missing value checks, drift detection)
9. Briefly describe an approach for testing/experimentation in production (e.g., shadow deployment, canary release, A/B testing) and one security consideration relevant to your ML system (e.g., input validation to prevent adversarial inputs, model/data access control).

Submission Guidelines

Prepare a Word/ PDF document containing the project details (All tasks, screenshots of the application as required, along with explanation) and code. Highlight clearly the contribution of each group member in executing the assignment. Upload the file as <groupid.docx> or <groupid.pdf> into the Taxila portal.

All the Best!